Sai Suddhir A B

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Education

Indian Institute of Technology Madras *('22 – '27)*

B.Tech in Aerospace Engineering + M.Tech in Electric Vehicles (Dual Degree) **CGPA:** 8.55/10

Mahindra World School, Chennai

Class XII (CBSE), 2022 95.8%

Class X (CBSE), 2020 96.16%

Conference Paper

- **Sai Suddhir A B** and Srikanthan Sridharan. "A Geometric Control Approach for DC-link Voltage Equalization in Split-Capacitor Inverters." submitted to *ICPE 2027* (under review).

Research and Professional Experience

Low-Cost Real-Time Simulator for Power Converters *Dec '25 – May '26*

Advisor: Prof. Lakshminarasamma N | Professor, IIT Madras

- Developed a real-time **boost-converter emulator** on a 570-LE Altera MAX V CPLD using Tustin-discretised Q16.16 fixed-point Verilog, capturing both CCM and DCM operations and tracking Typhoon HiL **within 1%**.
- Engineered a low-cost **CHIL** (Controller Hardware-in-Loop) testbed pairing the RTL plant solver with TI F28379D, running switching and averaged models in real time, validated against Typhoon HiL and PLECS.
- Designed a **controller interface card** with level shifting, analog signal conditioning and transient protection, enabling the F28379D controller to be validated on both Typhoon HIL-404 and the MAX V emulator.

Dual-Degree Project *May '26 – Present*

Advisor: Prof. Srikanthan Sridharan | Assistant Professor, IIT Madras

- Derived a **second-order switching surface** for split DC-link voltage balancing in a 3-level NPC DSTATCOM.
- Designing a **curve-fitting observer** to reconstruct three-phase VFD currents from a single DC-link sensor, fitting a sinusoidal oscillator to sparse, switching-state dependent samples without load parameters.

TVS Motor Company, Hosur *Jul '25*

Production Engineering Intern

- Engineered a **control algorithm** to retrofit conventional brake-fluid filling machines with **ABS filling** capability.
- Developed a closed-loop link between the bike's **hydraulic ECU** and the **PLC** over ISO-TP/CAN and OPCUA.
- Built a PLC-triggered **Python ISO-TP routine** that cycles the ABS modulator valves to purge trapped air.
- Projected to save **$200k+** in upgrade costs, earning commendation from VP (Production) and a full-time offer

Arctus Aerospace, Bangalore *Jun '25*

Hardware Engineering Intern

- Architected and built a modular **ground control station** for high-altitude UAVs, from system integration and wiring to enclosure, enabling manual and autonomous flight, real-time telemetry, FPV and in-flight tuning
- Designed **multilayer PCBs** and **STM32 firmware** for fuel, propulsion, braking, steering and battery systems.

Technical Projects

Avishkar Hyperloop, IIT Madras *Sep '23 – Nov '24*

Electrical Team Lead

- Designed a **6-layer dual-MCU board** housing TI's MSP432 and F28379D, executing v/f, SVPWM, inverter gate drive, regenerative braking, sensor fusion, intra-pod CAN telemetry and precharge for LIM propulsion.
- Developed **TI-RTOS PID firmware** for HV battery thermal control, holding within $\pm 0.5\,^{\circ}\mathrm{C}$ across cooling nodes.
- Architected a pod-wide **300+ message CANdb++ network** and a unified CAN.c driver with live ECU structs.
- Implemented a **20-state** fault-handling **finite state machine** in MATLAB Stateflow, covering **130+ fault cases**.
- Designed **multilayer PCBs** for propulsion, telecom, cooling, levitation, lighting, safety and braking systems.

Aero Club, IIT Madras *Sep '23 – Nov '24*

Embedded systems Engineer | Drone Swarm

- Developed a custom **STM32 flight controller** integrating telemetry, swarm algorithm and precise motor control, achieving a **70% hardware cost reduction** over a commercial Pixhawk.
- Modelled a **3-body anti-collision algorithm** in Python using NumPy and Shapely, with an **85% success rate**.

RTC-Based Street-Light Timer

Company: ROLFS, Chennai

- Designed an ATtiny85 **timer PCB** that reads an RTC over I^2C via TinyWireM library to drive relay switching.
- Kept it low-cost and user-friendly for **mass deployment**, now running **2000+ street lights** across Coimbatore.

Bidirectional Multi-Output DAB Converter for EVs *Jul '25 – Nov '25*

Course: Power Electronics & Motor Drives (ED5231)

- Designed and simulated a 500 V $\rightarrow$ 48 V/12 V, 100 kHz converter with **phase-shift modulation** and **inherent ZVS**, sizing the leakage inductance and selecting SiC MOSFETs and a planar transformer, achieving **96.2% efficiency**.

1 kW Half-Bridge Series Resonant Converter *Jan '26 – May '26*

Course: Switched Mode Power Conversion (EE5200)

- Designed a 400 V $\rightarrow$ 48 V, 100 kHz telecom DC-DC converter in PLECS, sizing the resonant tank by **FHA-based impedance modelling** and selecting switches and cores from conduction, switching and core **loss analysis**.

Real-Time Digital Control Firmware on TI C2000 *Jan '26 – May '26*

Course: Digital Controller for Power Applications (EE5212)

- Developed real-time **SPWM firmware** on the F28379D, configuring ePWM time-base, compare, action-qualifier and dead-band modules for phase shift, harmonic injection and active PFC, with synchronous ADC sampling.
- Designed a **boost controller** with hysteresis control, lag compensation, overcurrent protection and precharge.

Key Courses and Skills

Teaching Assistantship : Analog & Digital Electronics (ED2130) · Switched Mode Power Conversion (EE5200)

Key Courses :

Power Electronics	Power Converter Analysis & Design · Switched Mode Power Conversion · Power Electronics & Motor Drives · Digital Controller for Power Applications · Electric Machines · Battery Technology · EV Engineering & Development
Controls	Linear Dynamical Systems · Control of Automotive Systems · Signals & Systems
Mathematics	Probability, Statistics & Stochastic Processes · Complex Variables · Differential Equations · Series & Matrices · Functions of Several Variables

Technical Skills :

Programming	C, C++, Python (NumPy, PyTorch, TensorFlow), MATLAB, Verilog, VHDL
Software	PLECS, Simulink, LTspice, Altium Designer, KiCad, Code Composer Studio, STM32, Quartus, ModelSim, Vector CAN, PCAN, ROS2, Gazebo

Achievements

- Placed **1st in Asia** and **5th overall** at European Hyperloop Week 2024, Zurich, with Avishkar Hyperloop.
- Ranked in the **top 0.82%** in JEE Main and **top 0.41%** in JEE Advanced among ~1 million candidates.
- Delivered an **invited talk** on hyperloop technology to 200+ students at College of Engineering Trivandrum.
- Awarded the **Groww Foundation Scholarship** (2025–26) and the **Alumni MCM Scholarship** by IIT Madras.

Extra-Curricular Activities

- Served on the **Co-Curricular Affairs Executive Committee**, overseeing policy, budgets and conflict resolution.
- Competed at **Harvard MUN** and won special mentions at MVM and SRM MUN for a compelling resolution.
- Captained the **school cricket team** for two years and represented the school in inter-school athletics meet.
- Trained as a **Carnatic vocalist** and **keyboardist**, with keyboard certification from Trinity College, London.